

Lab#6: Knoxcraft Terps

In this lab, you will write code in VS Code that builds 3D structures to be viewed in Minecraft.

Write the code in VS Code.

Click the “Run” button to upload the code to our Minecraft server.

Log into Minecraft.

Run the program in

Download knoxcraft.zip

Download, unzip, and open knoxcraft in VS Code.

knoxcraft.zip is on Google Classroom.

VS Code

You will have programs like what we have below.

The `serverUrl` is correct

Set `minecraftPlayername` to your minecraft playername

Leave `username/password` the way they are

Change `programName` and `description` to reflect what you have done

```
import edu.knox.knoxcraftmod.client.*;

public class Mauritius {

    public static void main(String[] args)
    {
        String serverUrl = "http://54.226.75.11:8080";
        String minecraftPlayername = "spacdog";
        String username = "test";
        String password = "foobar123";
        String programName = "flag";
        String description = "Flag of Mauritius";
    }
}
```

```

        Terp terp = new Terp(programName, description);

        int length = 12;
        int width = 4;
        for (int i=0; i<width*4; i++) {
            for (int j=0; j<length; j++) {
                terp.forward();
                if (i / 4 == 0)
                    terp.setBlock(TerpBlockType.RED_WOOL);
                else if (i / 4 == 1)
                    terp.setBlock(TerpBlockType.BLUE_WOOL);
                else if (i / 4 == 2)
                    terp.setBlock(TerpBlockType.YELLOW_WOOL);
                else
                    terp.setBlock(TerpBlockType.GREEN_WOOL);
            }
            for (int j=0; j<length; j++) terp.back();
            terp.right();
        }

        TerpUploader.upload(serverUrl, terp, minecraftPlayername, username,
password);
    }
}

```

Terp turtle methods

You can run the following commands for a Terp turtle:

```

forward()
back()
left()
right()
up()
down()
turnLeft()

```

```
turnRight()  
setBlock(TerpBlockType)
```

The different `TerpBlockTypes` are listed here:

<https://jspacco.github.io/knoxcraftmod/javadoc/edu/knox/knoxcraftmod/client/TerpBlockType.html>

Use the run button in VS Code to upload to our Minecraft server

It will say "Upload Successful" if it works.

Minecraft slash commands

Once you are logged into Minecraft, you can run the following slash commands to summon your turtle, and to view and run your Java programs:

`/terp list` # list your Terp programs.

`/terp summon` # summon your Terp to your current position

`/terp forward|back|up|down|left|right|turnleft|turnright` # move or turn your terp — useful for adjust the start position of your program

`/terp run <program_name>` # run the program with program_name.

Here are instructions on how to run Minecraft

Write Java code to build at least **two** structures

- If you have ideas of what you want to build, do that!
- If you aren't sure, here are some ideas:
 - A house or building, which has a roof and walls
 - An abstract pattern, similar to the repeating patterns many of you did for Lab #2 Turtles, but with a third dimension
 - A three-dimensional version of one of the flags from Lab #3 Flagmaker that looks like it's waving

What to submit

- Zip and upload your project folder
- Include at least 2 **screenshots** of each structure